

ADR 0120 — A closed desk is proved twice, and never once on the wire

- **Status:** Accepted — implemented, mutation-proved (three arms run; two overlapped, a third found the genuinely disjoint second way), all caught
- **Date:** 2026-09-05
- **Issue:** #550 (M11.05)
- **Extends:** [ADR 0117](#) (the census and the fence this reads) · [ADR 0119](#) (`CensusPaths` , which this is the second caller of)
- **Supersedes / superseded by:** —

Context

M11.03 gave the drawer a door in (`/api/select-worktree`). #550 asks for the door out: `git worktree remove <path>` , addressed the same way — an opaque census id, never a path a client names. The design spec sketched the operation as `RemoveWorktree { path: PathBuf, id: WorktreeId }` and stated the rule in one sentence:

The stable `id` is the mutation authority. Under the repository guard, immediately before executing: take a fresh census, resolve `id` to its current path, and refuse unless that path equals the submitted path. Then remove by the resolved path — never by the submitted one.

That sentence assumes a client submits a path to compare against. Nothing in this codebase's wire contract may — `git-vista-protocol` is wasm-safe and carries no filesystem type at all, the same constraint that had already cut `AddWorktree`'s own `path` field (#656, unmerged as of this issue's base commit; read as reference, depended on as nothing). So the sentence has to be implemented without ever putting a path on the wire, and the "submitted path" it describes has to come from somewhere else.

Spec's sketch

RemoveWorktree{
 path, id }
 compare id's
 resolved path
 to the SUBMITTED
 path



The wall

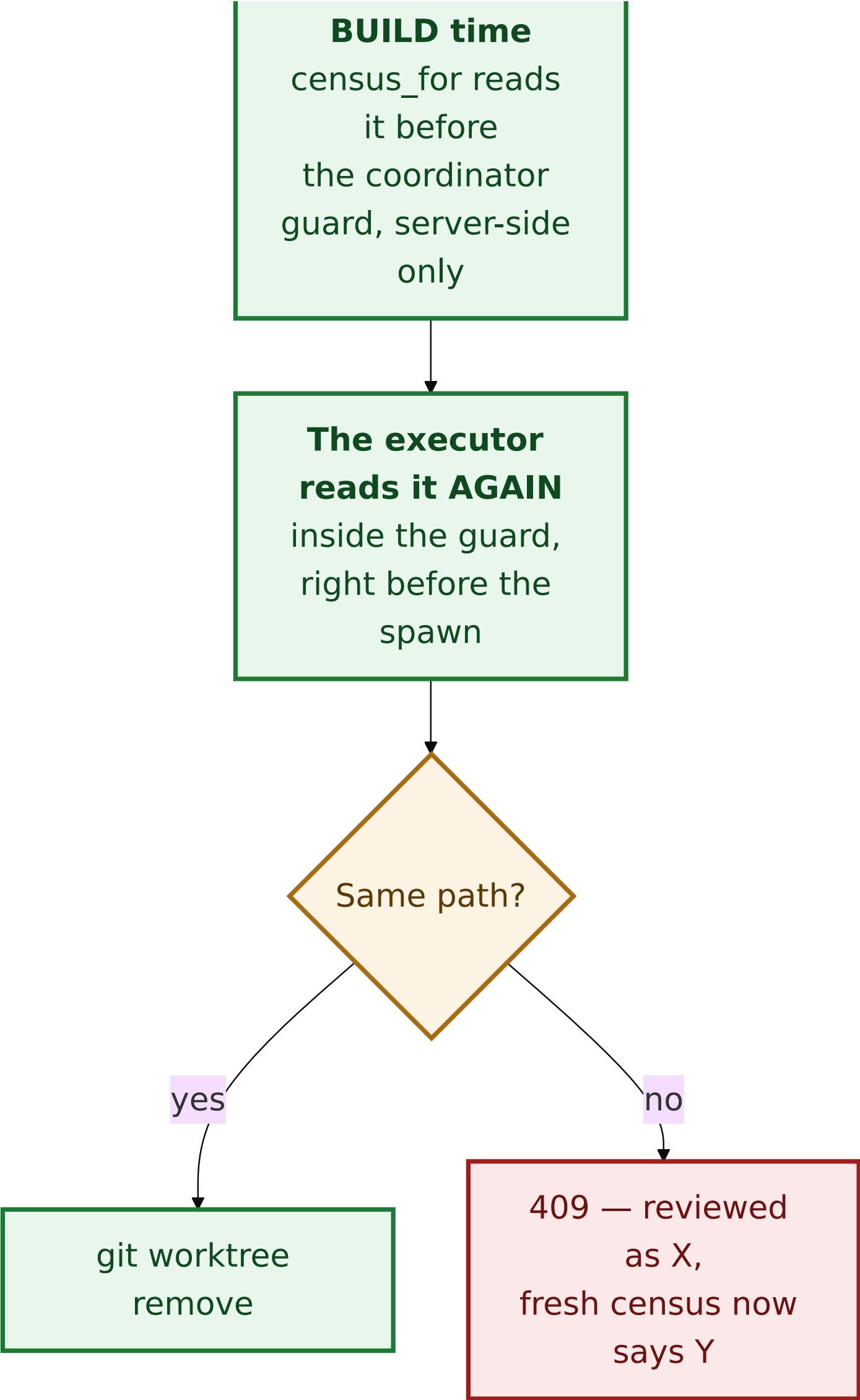
git-vista-protocol
is wasm-safe:
 no PathBuf,
 anywhere, ever



Where does the
'submitted path'
come from,
if not the wire?



Answer: plan-



Decision

1. `GitOperation::RemoveWorktree` carries `id` only

```
RemoveWorktree { id: WorktreeSiblingId }
```

`WorktreeSiblingId` is a new validated newtype (`require_non_empty`), kept distinct from `WorktreeToken` — which names the worktree a plan is built *against* — because the two are never interchangeable: removing the worktree you are standing in is refused server-side, never requested, and giving them separate types turns a mixed-up call into a compile error rather than a runtime refusal discovered later.

No `path` field, matching the correction #656 already made for `AddWorktree` for the identical reason. `RemoveWorktreeRequest` (the wire DTO) carries the same single field.

2. Two independent census reads stand in for "submitted vs. resolved"

The compare-and-swap the spec describes needs two values to compare. With no path on the wire, both have to be **server-derived reads of the same id, taken at two different times:**

- **The "expected" read** is `planner::census_for`'s ordinary plan-build pass — the same call `CheckoutBranch`'s collision precondition already pays for (M11.02, #547) — extended to also cover `RemoveWorktree`. It runs *before* the coordinator guard is acquired, exactly where every other plan's observation runs.
- **The "fresh" read** is a *second*, independent call to the same census function, made by `worktree_exec::exec_remove_worktree` itself, **inside** the coordinator guard `plan_and_execute_in` already holds for the whole `execute()` call. This is the read `enforce_fresh`'s own `observe_live` takes for every other precondition-bearing operation — except that value is computed and then discarded (never passed to `execute`), so a `Precondition` built on it could prove the id was *valid* but could

not hand the executor the *path* to actually remove. That is decision 3 below.

Both reads use `CensusPaths::rows_for_local_use` (ADR 0119): rows always carry the real path, regardless of the operator's `GIT_VISTA_EXPOSE_PATHS` setting, because neither read is ever serialized to a client — they exist purely for this internal comparison and the git spawn that follows it. This is the same constructor, and the same reasoning, `handlers::select::select_discovered_worktree` already uses for its own internal-only census read.

The comparison, and every refusal it can produce, lives in one function (`resolve_removable`) called against *both* reads, so the two are judged by the exact same rule:

case	refusal
census unreadable	409, the census's own client-safe reason
<code>id</code> names no sibling	404 — already gone
the sibling <i>is</i> the served worktree	409 — defense in depth; the drawer never offers this
<code>Serviceable::Missing</code>	409, <code>Serviceable::refusal()</code> 's own words — releasing it is <code>git worktree prune</code> , a different operation this design omits
<code>Serviceable::OutsideAllowedRoots</code>	409, <code>Serviceable::refusal()</code> 's own words — visible for collision detection only; visibility must never widen the mutation boundary
expected path $\neq$ fresh path	409 — "this worktree changed since it was reviewed"

Only on a match does `git worktree remove <fresh path>` run — the **fresh** path, never the expected one, matching the spec's own instruction to remove by the resolved value and not the submitted one.

3. No `Precondition` — the gate lives in the executor, by name

Every other destructive operation in this vocabulary states its gate as a `Precondition`, re-verified by `enforce_fresh` before `execute` ever runs. This one does not, and the reason is structural rather than stylistic: `enforce_fresh` re-derives a fresh `Observed` (`observe_live`) purely to *check* it against the plan's preconditions — the fresh value itself is thrown away the moment the check passes. A `Precondition::WorktreeStillAt` could prove "the id still resolves," but it could not hand `execute` the path that resolution produced, because nothing propagates `enforce_fresh`'s internal `live` past its own return.

`shape()` therefore gives `RemoveWorktree` **no preconditions at all** — `RiskLevel::Destructive`, `RecoveryStrategy::Irrecoverable`, empty preconditions, no ref changes — and the whole compare-and-swap lives in `exec_remove_worktree`, which takes its own second census read specifically because it is the one place that can use the path it just proved. This is recorded as a deliberate choice in both `GitOperation::RemoveWorktree`'s doc comment and `shape`'s own arm, so a future reader does not "fix" the missing precondition by adding one that cannot carry what it would need to.

4. `git worktree remove` needs a grant it does not have, and the grant is proved, not assumed

`git worktree remove <path>` writes outside the served repository — into the sibling's own directory — and `sandbox::policy_for` never grants that: it grants the served repository (and its `common_dir`, for a linked worktree) and the fixed system trees, nothing else.

`sandboxed_with_extra_grant` / `git_output_with_extra_grant` (new, `git_cmd.rs`) compose one extra RW grant on top of the ordinary policy — same shape as `AddWorktree`'s unmerged `sandboxed_with_grant`, kept as a separate name deliberately (a future merge of #656 should deduplicate the two, not assume either is gone).

The grant target was wrong on the first attempt, and the fix was found by this feature's own pipeline test, not reasoned out in advance. Granting `fresh_path` itself is not enough: `git worktree remove` deletes everything *inside* the directory (which a grant on the directory covers) and then unlinks the directory entry from its **parent**, which needs the parent writable. The first version of `exec_remove_worktree` failed `remove_worktree_executes_through_the_pipeline` with a real `Permission denied` on that last step; the fix is `fresh_path.parent()`, not `fresh_path`. Left in as the concrete demonstration of why "run the real thing through a real sandbox" is not optional even for a feature whose logic looks obviously right on paper.

The grant is never client-derived: it is `fresh_path`'s parent, and `fresh_path` is a value this very call just proved, via a live census, to be `Serviceable::Yes` — already inside this application's own allowed roots.

5. `--force` is not offered, anywhere, and that is checked structurally

Git's own refusal on a dirty tree is the entire protection an uncommitted, never-staged edit in the removed worktree has — it was never written to git's object database, so there is nothing for this app's recovery machinery to pin.

`remove_worktree_never_passes_force` greps `worktree_exec.rs` for the literal, quoted `"--force"` (not the bare substring, which this file's own doc comments legitimately use in backtick prose) — the same posture `argv_boundary`'s tripwires already take for other structural absences.

6. The two-tap ceremony gets its own function, not a third `WorktreeAction` arm

`features::dialogs::core::WorktreeAction` (`DiscardTracked / DeleteUntracked`, #220) addresses tracked/untracked **paths** inside the served worktree. `RemoveWorktree` addresses a whole **other** worktree by id — a different subject with no path list to render and no per-path recovery nuance to word.

`remove_worktree_confirm(name, armed)` is a new, parallel function with the same two-tap `ArmStep` shape `DeleteUntracked` already established, wired into `dialogs/confirm.rs`'s existing `PendingOp` match and `OperationKind`'s existing dispatch table (`write_route`, `send()`) — no parallel mechanism, per the issue's own instruction.

The drawer's row gains a **second, independent offer** (`RemoveOffer`, alongside the existing `RowOffer`), not a third `RowOffer` variant: a servable, non-current row answers yes to both "can I switch to this?" and "can I close this?" at once, which one offer field per row cannot express.

Alternatives considered

- **Carry `path` as a `String` on the wire, client-supplied.** Rejected: every other request body in this codebase addresses by opaque id specifically so a location on disk is never something a request names, and a client has no legitimate way to have learned a real path (`GIT_VISTA_EXPOSE_PATHS` is off by default).
- **A `Precondition::WorktreeSiblingResolves` re-verified by `enforce_fresh`.** Rejected — see decision 3: the fresh value `enforce_fresh` computes to check a precondition is discarded before `execute` runs, so this shape cannot hand the executor the path it needs without a larger, riskier change to shared pipeline plumbing that no other operation needs.
- **Reuse #656's `sandboxed_with_grant` verbatim.** Not possible: #656 is unmerged, and depending on unmerged code would make this issue's own landing conditional on another lane's. A near-identical function was written under a different name, with a note to deduplicate at merge time.
- **Grant `fresh_path` (not its parent).** Tried first; refuted by the pipeline test itself (decision 4) rather than reasoned out before writing code.

Consequences

- One new POST route (`/api/remove-worktree`), classified in `ROUTE_AUTHZ` (`SessionAndCsrft`) and in the planner's write-route census as a genuine git write, with its own funnel row.
- A new crate-internal sandbox primitive (`sandboxed_with_extra_grant`) that the eventual `AddWorktree` merge should fold together with its own `sandboxed_with_grant` rather than carry both.
- `features::operations::core,signals.rs` 's exhaustive `OperationKind` census (`write_route` , `every_operation_kind` , `sends_dispatch_matches_the_route_table`) now has a fourteenth entry — every one of those censuses is compile- or test-enforced, so a future variant cannot land partially wired.
- Explain Mode needed **zero new arms**: `RiskLevel::Destructive` and `RecoveryStrategy::Irrecoverable` are already-covered facts, proved by building a real plan and reading every rendered line (`a_remove_worktree_plan_explains_itself_with_no_new_sentence`), the same proof style #656 established for `AddWorktree` .
- The checklist export (`plan_export.rs`) refuses to print a copyable command line for this operation, for the same reason it refuses one for `AddWorktree` : the real argument (a resolved path) is not something the client — or a person reading the checklist — can know in advance.

Mutation proof

Three arms against `crates/git-vista-server/src/planner/worktree_exec.rs` , proved via `failure-atlas` 's `mutation_check` (a fresh clone at HEAD, run unmutated then mutated, never touching this working tree), all **caught** by `remove_worktree_refuses_when_the_id_now_resolves_elsewhere` — the test that drives the exact scenario the spec names: the reviewed desk is closed and a different one reoccupies the same admin-dir slot, so `id` is unchanged but the real path is not.

arm	mutation	mutated result
remove the check	delete <code>if fresh_path != expected_path { ... } (let _ = &expected_path;);</code> both census reads still run, the fresh resolution is just never compared	200 Worktree removed where <code>409</code> was expected — the operation actually ran, silently redirected onto whatever now occupies <code>id</code>
invert the comparison	<code>!=</code> → <code>==</code> (refuse on a match, proceed on a mismatch — backwards)	200 Worktree removed — same symptom as the first arm
weaken to id-existence-only	keep the fresh census read but drop both the <code>Serviceable</code> /path resolution AND the equality check: only confirm <i>some</i> sibling with this <code>id</code> still appears in the fresh census (<code>siblings.iter().any(s s.id == id)</code>), then act on the stale, build-time path rather than re-deriving it	400 Bad Request — <code>fatal: '<stale path>' is not a working tree</code> , because the reviewed desk had genuinely been removed and the weakened check never re-derived where <code>id</code> lives now

The first two initially looked like one proof run twice — both go red on the identical `200` vs. `409` symptom, because both leave the *fresh* path resolution intact and only break the *comparison* against it. The third arm is the one that actually earns "two ways": it breaks a different half of the mechanism (the re-derivation itself, not the comparison) and is caught by a **different** failure mode — git's own error surfacing as `400`, rather than the application silently succeeding as `200`. That the first two overlap is recorded here rather than quietly dropped; the third is what makes the proof honest. All three: caught, none survived. Run ids: 280 (delete the check), 281 (invert the comparison, redundant with 280 — see

above), 282 (weaken to id-existence-only); run_key: gv-550-
remove-worktree-cas .